SAI NAVEEN CHANUMOLU

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SUMMARY

Software Engineer with 3 years of experience in designing and delivering scalable enterprise applications, cloud-native microservices, and AI-integrated solutions. Skilled in full-stack development with expertise across Java, Python, TypeScript, React.js, Angular, Spring Boot, Node.js, and Flask, complemented by strong knowledge of databases, DevOps, and cloud platforms (AWS, Azure, GCP). Adept at applying Agile, TDD/BDD, and CI/CD practices to produce high-quality, maintainable software solutions. Demonstrated success in solving complex challenges involving system design, distributed systems, and performance optimization, while collaborating seamlessly with global teams. Passionate about leveraging emerging technologies such as LLMs, MLOps, and event-driven architectures to drive innovation and operational efficiency.

SKILLS

- **Programming:** Java, Python, TypeScript, C, C++, JavaScript, SQL, .Net
- **Frontend:** React.js, Angular, HTML5, CSS3, Tailwind CSS
- **Backend:** Spring Boot, Node.js, Express.js, Django, Flask, FastAPI
- **Databases:** MySQL, PostgreSQL, MongoDB, SQL Server, Redis
- **DevOps & Cloud:** AWS, Azure, GCP, Docker, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions
- **Messaging & APIs:** Kafka, RabbitMQ, REST, GraphQL, SOAP, Swagger
- **Testing & QA:** JUnit, Jest, Mocha, Cypress, Selenium
- **Security:** OAuth2, JWT, SSL/TLS, RBAC
- **AI & Intelligent Systems:** TensorFlow, PyTorch, Scikit-learn, OpenAI API, Hugging Face Transformers, LLM Integration, Pandas, NumPy, MLOps (model deployment & monitoring)
- **SDE Concepts:** Data Structures & Algorithms, Object-Oriented Design (OOD), Design Patterns, System Design, Distributed Systems, Microservices, Multithreading & Concurrency, Performance Optimization, Problem Solving, Clean Code
- **Methodologies & Tools:** Agile (Scrum, Kanban), TDD, BDD, Git, Jira, Confluence, SonarQube

WORK EXPERIENCE

Honeywell — **Jun 2025 –Present**

Software Engineer

- Driving the Honeywell Forge Platform Enhancements project by developing cloud-native microservices that expand IoT data analytics and predictive maintenance capabilities.
- Built event-driven architectures using Node.js, FastAPI, and Kafka, enabling real-time anomaly detection and intelligent alerts for industrial operations.
- Deployed scalable applications on AWS with Kubernetes and Terraform, ensuring high availability and fault tolerance across global deployments.
- Integrated LLM-based insights via Hugging Face and OpenAI APIs to enhance operational decision-making and boost worker productivity.
- Implemented DevSecOps pipelines with GitHub Actions and SonarQube, achieving faster release cycles while maintaining code quality and compliance.

Deloitte — **May 2022 – Jul 2023**

Software Engineer

- Designed and integrated a new dynamic homepage for an employee performance management system, improving accessibility and reducing user task time by 90% using HTML, CSS, Bootstrap, TypeScript, and Angular.
- Revamped project UI/UX for responsiveness and optimized .NET application performance by 15%.
- Developed a scalable course recommendation system using Node.js and Flask microservices, replacing a legacy vendor tool and improving maintainability.
- Built and deployed RESTful APIs in Node.js and Flask to serve React-based frontend clients, ensuring modularity, testability, and scalability.
- Enhanced recommendation accuracy by 15% through a collaborative filtering algorithm, leveraging user behavior and course completion data for personalized insights.
- Led migration of HR data workflows after restructuring the core database, redesigning schemas to consolidate roles, synchronize employee details, and enable seamless integration of new hires.
- Refactored stored procedures and optimized complex SQL queries (joins, subqueries, nested queries), reducing processing time by 20% and improving data accuracy.

- Conducted impact analysis and refactored code across frontend and backend systems, ensuring compatibility during core data source migration and managing REST APIs.
- Participated in company hackathons, developing machine learning POCs and integrating them into internal applications to solve business challenges.

Trigent **Nov 2020 – Apr 2022**
Software Engineer

- Contributed to the Global Insurer GCC Modernization project by developing scalable modules that migrated legacy policy administration systems into microservices architecture.
- Engineered backend services using Java, Spring Boot, ensuring high availability and seamless data integration across enterprise platforms.
- Optimized database interactions in SQL Server and MySQL, reducing query execution time and improving data retrieval efficiency for critical insurance workflows.
- Designed and implemented RESTful APIs to connect customer-facing portals with backend systems, improving cross-platform interoperability.
- Automated regression and functional testing with JUnit and Selenium, accelerating release cycles and reducing defect leakage by 40%.
- Collaborated with cross-functional teams in an Agile Scrum environment, delivering sprint-based features and ensuring alignment with client requirements.
- Applied design patterns and object-oriented principles to produce clean, maintainable, and reusable code across core system modules.
- Partnered with onshore/offshore teams to ensure compliance with security standards such as OAuth2 and TLS**,** safeguarding sensitive policyholder data.

PROJECTS

Matra AI – HackDavis

- Developed a full-scale application to help bilingual children improve English speaking skills by listening and reading stories with our AI, which nurtures the child like mother by providing feedback and correcting their pronunciation.
- Built with React (frontend), Flask (backend), and integrated OpenAI, Speechace, and ElevenLabs APIs for speech to text and text to speech functionality. Deployed on Heroku and accessible via a custom .Tech domain.
- Won Best Interactive Media Award at Hack Davis hackathon conducted at UC Davis with 900+ participants.

Smart Traffic Flow Analyzer

- Designed a data-driven system to analyse real-time traffic datasets using Python, Pandas, NumPy, and Scikit-learn for congestion prediction and route optimization.
- Implemented regression and clustering models to forecast traffic density trends as part of data analytics coursework.
- Built an interactive dashboard with Tableau to visualize traffic insights and support urban planning case studies.

Automated Student Attendance & Report Generator

- Developed an automation tool in Python with OpenCV for face recognition–based attendance tracking in academic settings.
- Integrated attendance logs with a MySQL database and automated report generation in Excel and PDF formats using SQL and Python scripts.
- Delivered a functional prototype that showcased automation and integration for streamlining classroom administrative tasks.

EDUCATION

Master in Software Engineering **Aug 2023 – May 2025**
San Jose State University – San Jose, California | GPA: 3.57